GOV•UK

Guidance

Using chatbots and webchat tools

How to use chatbots and webchat tools to improve your users' experience of your service.

From: **Government Digital Service (/government/organisations/government-digital-service)** and **Central Digital and Data Office (/government/organisations/central-digital-and-data-office)**

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Contents

— Understanding the difference between chatbots and webchat

— Setting up your project

— Set expectations early

— Build a knowledge base

— Build conversations

— Help users reach their goals

— Provide alternatives to the tool

Chatbots and webchat systems are web tools that can diversify how your users make contact with your organisation. They can help users find the information they need or get help in completing tasks in an alternative way to what your organisation currently offers.

This guidance provides a set of principles to consider when designing a chatbot or webchat system. It also includes case studies from departments who have implemented or experimented with these tools.

Understanding the difference between chatbots and webchat

A webchat (also known as live chat) is an online conversation between a user and a human advisor. Webchat usually requires a trained team of advisors to handle user inquiries.

A chatbot is an online application that can help users solve problems without a human advisor. They are usually programmed to recognise phrases or words and use them to provide answers or links to information. They are sometimes known as a virtual assistant (VA).

There are different types of chatbots:

- menu-based chatbots that follow a closed script so users can choose from predefined options
- keyword recognition chatbots that read and interpret key words and respond to typed enquiries
- advanced chatbots that use Natural Language Processing (https://dataingovernment.blog.gov.uk/2019/06/14/natural-language-processing-in-government/) (NLP) capabilities and can learn from continuous user interactions

You can also find chatbots that combine these 3 elements. For example, a chatbot that uses both keyword recognition and a menu structure. Chatbots can have different use cases such as customer support, automating repetitive tasks or as an alternative content format. For example, the Office of National Statistics (ONS) created bot to provide information on job automation (https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/employmentandemployeetypes/articles/whichoccupationsareathighestriskofbeingautomated/2019-03-25) and the Ministry of Justice (MoJ) explored users' experience of website content compared to using a chatbot. (https://www.gov.uk/government/case-studies/how-the-ministry-of-justice-explored-using-a-chatbot)

Similar interfaces, but different technologies

Chatbots and webchat use different technologies and processes. Users might see chatbots and webchat as similar services, so it's important to make

it clear to users which type of tool they're using. Your tool should complement your existing contact services and not be the only way for users to make contact or find help.

Setting up your project

To plan your project, you should use the Service Standard (https://www.gov.uk/service-manual/service-standard) and conduct user research (https://www.gov.uk/service-manual/user-research) to find out which type of tool is best for your users and organisation. In some cases, you might find it more time and cost-effective to improve existing content, navigation and the search functionality on your website.

During planning you should consider:

- your users' needs
- the issues you aim to solve
- whether the chosen tool meets your organisation's business needs

You must also comply with the General Data Protection Regulation (https://www.gov.uk/government/publications/guide-to-the-general-data-protection-regulation) (GDPR) if you are storing users' personal data. The Information Commissioner's Office has more information (https://ico.org.uk/for-organisations/guide-to-data-protection/guide-to-the-general-data-protection-regulation-gdpr/) about how to protect users' personal data and comply with GDPR.

You should consider some simple questions when choosing what tool you need for your service.

1. How will this tool fit into our existing service?
2. What can the tool do and what are its limitations?
3. What information does the user need to provide?
4. What type of response or decision will the user need?

Follow the Technology Code of Practice (https://www.gov.uk/government/publications/technology-code-of-practice/technology-code-of-practice#use-cloud-first) to help you decide whether to build or buy a chatbot or webchat tool. You can find suppliers through the Digital Marketplace (https://www.digitalmarketplace.service.gov.uk/).

To future-proof your service, outline the capabilities you want now, but also make sure the service is flexible enough for future changes. For example,

consider how well the tool can integrate into your business processes. This will help you plan and manage how your business processes will need to change to incorporate the new tool and its maintenance.

Set expectations early

You should be clear about what tools you're providing for your users. Sometimes, users struggle to spot the difference between chatbots and webchat and might get confused or frustrated.

For chatbots, it's best to make it clear the user is not talking to a real person. For example, include a statement that they are using an automated service and avoid using a fictional name or an image of a person. For both types of tools, you should use the GOV.UK Design System (https://design-system.service.gov.uk/patterns/contact-a-department-or-service-team/) to present a clear and consistent visual design.

Outline what your tool can and cannot do on a landing page or at the start of your users' chat session. For example, explaining what help the users can expect and what kind of questions they can ask. For Natural Language Processing (NLP) chatbots, it is useful to show users an example of how to phrase questions. This will improve the results from the chatbot.

Build a knowledge base

Creating a knowledge base can help you to categorise user requests and improve the value of your tool. Having information organised in a clear and structured format can help your webchat advisors or the chatbot respond faster and more accurately.

You can build a content knowledge base by looking at:

- previous user telephone inquiries, emails or chat logs
- most commonly raised concerns
- analytics for the most searched and visited webpages
- feedback from users, advisors and team members

Building a good knowledge base takes time, but it allows you to expand the capabilities of your tool, find gaps in your content and improve your service.

For webchat, you might store a knowledge base in a customer relationship management tool or your organisation's intranet. Your chosen storage option

must have good search functionality so that webchat advisors can quickly find previous user conversations. If advisors handle more than one chat at once, they will need the option of looking up multiple records at the same time.

Building a knowledge base for NLP chatbots

If you want to build a chatbot that learns from Natural Language Processing (NLP) you'll need structured data so the bot can filter and extract the right information from your knowledge base. You need to classify your data with intents and entities (https://chatbotsmagazine.com/chatbot-vocabulary-10-chatbot-terms-you-need-to-know-3911b1ef31b4) so that it matches with your users' requests. The more intents you tag in your data, the more queries a NLP chatbot can answer.

The way in which you organise your data depends on the software you use and the type of data your chatbot needs. This can be the most time consuming part of building and training your NLP chatbot. NHS Digital talks about how they built a metadata database to support their ViDA chatbot project.

For your data to be useful, you need to train your bot to understand the different ways your users will express their intentions and goals. To do this you need to build a set of utterances (https://chatbotsmagazine.com/chatbot-vocabulary-10-chatbot-terms-you-need-to-know-3911b1ef31b4), which is the various ways a user can say the same thing, so that your bot can recognise the appropriate response.

Once you've trained the bot, it should be able to classify the users' requests even though the wording will vary. If the bot responds with a high level of accuracy the users will have a better experience. You should user test the response accuracy before your bot goes live.

Once live, the bot's accuracy might drop as it becomes exposed to requests beyond its scope. You can use this information to add more utterances and user intents to your dataset, which will help improve your bot.

Build conversations

Users often use tools like webchat or chatbots when they need support. Conversation patterns are different to writing web pages and users expect a level of reassurance and guidance specific to their enquiries. Avoid providing large amounts of information all at once and keep the information relevant.

A good conversation style for both tools should include:

- keeping text simple, concise and easy (https://www.gov.uk/guidance/content-design/writing-for-gov-uk) to follow
- asking questions to keep users engaged and feel supported during the session
- taking turns during the conversation and giving the user time to respond
- listening cues to give users confidence that the tool has correctly interpreted their message

An example of a listening cue.

> “User: I want to know how to appeal a decision
>
> Tool: Ok, I’ll fetch some data on the appeal process for you”

For webchat, you can have scripted responses to frequently asked questions to help advisors search for information and respond to users faster. This helps advisors deliver consistent advice to users, but leaves them the option to tailor their responses.

For chatbots, it’s important you map out potential user intents to avoid dead-ends in a user’s journey. Your bot should tell users when it needs them to rephrase their question. During planning, you should work out the questions that might lead to dead-end conversations and decide what responses you’ll use to unblock them. You can gather this data from analysing user research, call centre conversations, using existing data from website analytics or common issues raised by users.

Your users might benefit from testing with different formats, for example:

- a free-text box so users can write in their own words
- suggesting buttons or widgets to help users choose options quickly without typing

Help users reach their goals

Users want to take as few steps as possible to reach their goals. Their trust in your tool might disappear if they do not get a good service or the tool cannot help them.

Some ways you can improve your service include:

- providing proof of chat and transcripts
- optimising the placement of the tool in the service
- starting with simple and focussed transactions
- making your tool as accessible and inclusive as possible

Proof of chat and transcripts

Users need the option to refer back to conversations with your tool. You can meet this need by providing the ability to download transcripts or email transcripts to users. You should let users know about your chosen option before they begin a session. Clearly display where they can submit their email or include a visible download tab in the chat window.

Placement of the tool

Where you place your tool can make a big difference to how it's used. The decision on where it's placed will depend on:

- when and where users need support
- what your website analytics tell you about your current service
- where your users normally contact you

Your tool should be easy to find, but you do not want it to overshadow existing content or become the main focus of your service. Poor placement can lead to:

- users not finding your tool
- users not finding the information they need
- an increase in user dissatisfaction
- an increase in users phoning for help rather than finding information through the tool

Regular user testing will help you find what works best for your users.

Depending on the purpose of your tool, you can place the link on your main contact page or on the service pages. For example, HMRC has a different webchat link on every service (https://www.gov.uk/contact-hmrc) such as VAT, Child Benefit or Income Tax. After completing some security steps, users can speak to the appropriate webchat specialist for their inquiries.

Starting with simple and focused transactions

Start small and do a gradual roll out of your tool's scope and depth. This can help make sure your project does not lose focus by trying to do too much too quickly. This approach can also help your project collect more accurate and in depth feedback which is useful for future iterations. For example, DVLA rolled back a complicated version of their bot and did smaller, simpler iterations (https://www.gov.uk/government/case-studies/how-the-dvla-deployed-a-chatbot-and-improved-customer-engagement).

Make your tool accessible and inclusive

It is important to make your tool accessible and inclusive for all users. It is also required by law (https://www.gov.uk/guidance/accessibility-requirements-for-public-sector-websites-and-apps). You can improve your tool by using patterns and design methods (https://accessibility.blog.gov.uk/2016/12/09/patterns-for-accessible-webchats/) and including accessibility in your tool's design (https://www.gov.uk/service-manual/helping-people-to-use-your-service).

Provide alternatives to the tool

Your tool should be useful, not an obstacle to users getting the information they need. With chatbots, users can sometimes find themselves in a conversation loop. You can make sure this does not happen by providing a way to transfer to a real person, or include an option to escalate their request through webchat or a telephone call.

Users also need to have access to alternative options if they cannot find the support they need on your chatbot or webchat service. Having multiple options for contact can also improve the accessibility and inclusivity of your service.

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